WP9 Solutions

217. $w=5x^2+2y^2$, $x=-3s+t$, $y=s-4t$, $\frac{\partial w}{\partial s}$ and $\frac{\partial w}{\partial t}$

$$\frac{\partial w}{\partial s}=\frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial s}+\frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial s}$$
$$=10x(-3)+4y(1)$$
$$=-30x+4y$$

$$\frac{\partial w}{\partial t}=\frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial t}+\frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial t}$$
$$=10x(1)+4y(-4)$$
$$=10x-16y$$

223. $f(x,y)=xy$, $x=1-\sqrt{t}$, $y=1+\sqrt{t}$

$f_x=y \qquad f_y=x$

$x_t=\frac{-1}{2\sqrt{t}} \qquad y_t=\frac{1}{2\sqrt{t}}$

$$\frac{\partial f}{\partial t}=f_x\cdot x_t+f_y\cdot y_t$$
$$=(1+\sqrt{t})\left(-\frac{1}{2\sqrt{t}}\right)+(1-\sqrt{t})\left(\frac{1}{2\sqrt{t}}\right)$$
$$=-\frac{1}{2\sqrt{t}}-\frac{1}{2}+\frac{1}{2\sqrt{t}}-\frac{1}{2}$$
$$=-1$$

243. $z=\frac{x}{y}$, $x=2\cos u$, $y=3\sin v$, $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$

$$\frac{\partial z}{\partial u}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u}$$
$$=\frac{1}{y}(-2\sin u)$$
$$=\frac{-2\sin u}{3\sin v}$$

$$\frac{\partial z}{\partial v}=\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial v}$$
$$=-\frac{x}{y^2}(3\cos v)$$
$$=\frac{-2\cos u\cos v}{3\sin^2 v}$$

245. $z=xye^{x/y}$, $x=r\cos\theta$, $y=r\sin\theta$ $\frac{\partial z}{\partial r}$, $\frac{\partial z}{\partial \theta}$, $r=2$, $\theta=\pi/6$

$$\frac{\partial z}{\partial r}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial r}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial r}$$
$$=\left(ye^{x/y}+xy\cdot\frac{1}{y}e^{x/y}\right)\cos\theta$$
$$+\left(xe^{x/y}+xy\left(-\frac{x}{y^2}\right)e^{x/y}\right)\sin\theta$$
$$\left.\frac{\partial z}{\partial r}\right|_{(2,\pi/6)}=\left(e^{\sqrt{3}}+\sqrt{3}e^{\sqrt{3}}\right)\frac{\sqrt{3}}{2}$$
$$+\left(\sqrt{3}e^{\sqrt{3}}+\sqrt{3}(-\sqrt{3})e^{\sqrt{3}}\right)\frac{1}{2}$$
$$=\sqrt{3}e^{\sqrt{3}}$$

$$\frac{\partial z}{\partial \theta}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial \theta}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial \theta}$$
$$=\left(ye^{x/y}+xy\frac{1}{y}e^{x/y}\right)(-r\sin\theta)$$
$$+\left(xe^{x/y}+xy\left(-\frac{x}{y^2}\right)e^{x/y}\right)(r\cos\theta)$$
$$\left.\frac{\partial z}{\partial \theta}\right|_{(2,\pi/6)}=\left(e^{\sqrt{3}}+\sqrt{3}e^{\sqrt{3}}\right)(-1)$$
$$+\left(\sqrt{3}e^{\sqrt{3}}+\sqrt{3}\left(-\sqrt{3}e^{\sqrt{3}}\right)\right)\sqrt{3}$$
$$=(2-4\sqrt{3})e^{\sqrt{3}}$$

251. $V(x,y) = \pi x^2 y$, $x(t) = \frac{1}{2}t$, $y(t) = \frac{1}{3}t$

$$\frac{\partial V}{\partial t} = \frac{\partial V}{\partial x}\cdot\frac{\partial x}{\partial t} + \frac{\partial V}{\partial y}\cdot\frac{\partial y}{\partial t}$$
$$= (2\pi xy)\left(\tfrac{1}{2}\right) + (\pi x^2)\left(\tfrac{1}{3}\right)$$
$$= \pi xy + \tfrac{1}{3}\pi x^2$$
$$\left.\frac{\partial V}{\partial t}\right|_{(2,4/3)} = \pi(2)(4/3) + \tfrac{1}{3}\pi(2)^2$$
$$= \frac{8\pi}{3} + \frac{4\pi}{3}$$
$$= 4\pi$$

257. $x = \sqrt{1+t}$, $y = 2 + \frac{1}{3}t$

$T_x(2,3) = 4$, $T_y(2,3) = 3$

$$\frac{\partial T}{\partial t} = \frac{\partial T}{\partial x}\cdot\frac{\partial x}{\partial t} + \frac{\partial T}{\partial y}\cdot\frac{\partial y}{\partial t}$$
$$= T_x\left(\frac{1}{2\sqrt{1+t}}\right) + T_y\cdot\frac{1}{3}$$
$$\left.\frac{\partial T}{\partial t}\right|_{t=3} = 4\left(\frac{1}{2\sqrt{4}}\right) + 3\left(\tfrac{1}{3}\right)$$
$$= 1 + 1$$
$$= 2^\circ/\text{sec}$$

263. $f(x,y) = xy$; $P(0,-2)$; $V = \frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}$

$f_x = y \qquad f_x(0,-2) = -2$

$f_y = x \qquad f_y(0,-2) = 0$

$$D_u f(0,-2) = -2\left(\tfrac{1}{2}\right) + 0\left(\tfrac{\sqrt{3}}{2}\right)$$
$$= -1 + 0$$
$$= -1$$

269. $f(x,y) = e^x \cos y$; $u = \langle 0,1\rangle$; $P(0,\frac{\pi}{2})$

$f_x = e^x \cos y \qquad f_x(0,\tfrac{\pi}{2}) = e^0 \cos\tfrac{\pi}{2} = 0$

$f_y = -e^x \sin y \qquad f_y(0,\tfrac{\pi}{2}) = -e^0 \sin\tfrac{\pi}{2} = -1$

$$D_u f(0,\tfrac{\pi}{2}) = 0(0) + (-1)(1)$$
$$= 0 - 1$$
$$= -1$$

281. $f(x,y,z) = xy + yz + xz$ at $P(1,2,3)$

$f_x = y + z \qquad f_x(1,2,3) = 2+3 = 5$

$f_y = x + z \qquad f_y(1,2,3) = 1+3 = 4$

$f_z = x + y \qquad f_z(1,2,3) = 1+2 = 3$

$$\nabla f = 5\hat{i} + 4\hat{j} + 3\hat{k}$$

291. $f(x,y) = xe^y - \ln(x)$; $P(-3,0)$

$f_x = e^y - \frac{1}{x} \qquad f_x(-3,0) = 1 + \frac{1}{3}$

$= 4/3$

$f_y = xe^y \qquad f_y(-3,0) = -3e^0$

$= -3$

$$\nabla f = \tfrac{4}{3}\hat{i} - 3\hat{j}$$

307. $V(x,y,z) = 5x^2 - 3xy + xyz$

a) $P(3,4,5)$; $u = \langle 1,1,-1 \rangle$

$V_x = 10x - 3y + yz$

$V_x(3,4,5) = 30 - 12 + 20$

$= 38$

$V_y = -3x + xz$

$V_y(3,4,5) = -9 + 15$

$= 6$

$V_z = xy$

$V_z(3,4,5) = 3(4)$

$= 12$

$\frac{u}{\|u\|} = \frac{\langle 1,1,-1 \rangle}{\sqrt{1^2+1^2+(-1)^2}}$

$= \frac{\sqrt{3}}{3}\langle 1,1,-1 \rangle$

$D_u V(3,4,5) = 38\left(\frac{\sqrt{3}}{3}\right) + 6\left(\frac{\sqrt{3}}{3}\right) + 12\left(-\frac{\sqrt{3}}{3}\right)$

$= \frac{32\sqrt{3}}{3}$

b) which direction does the voltage change most rapidly?

from part a: $V_x(3,4,5) = 38$

$V_y(3,4,5) = 6$

$V_z(3,4,5) = 12$

$\nabla V = 38\hat{i} + 6\hat{j} + 12\hat{k}$

c) max rate at that point

$\|\nabla V\| = \sqrt{38^2 + 6^2 + 12^2}$

$= \sqrt{1624}$

$= 2\sqrt{406}$.